

Tutorial 02

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Example 1.4.6 – Using a Graph to Represent Knowledge

The directed graph shown in Figure 1.4.2 is a pictorial representation for a simplified knowledge base about periodical publications.

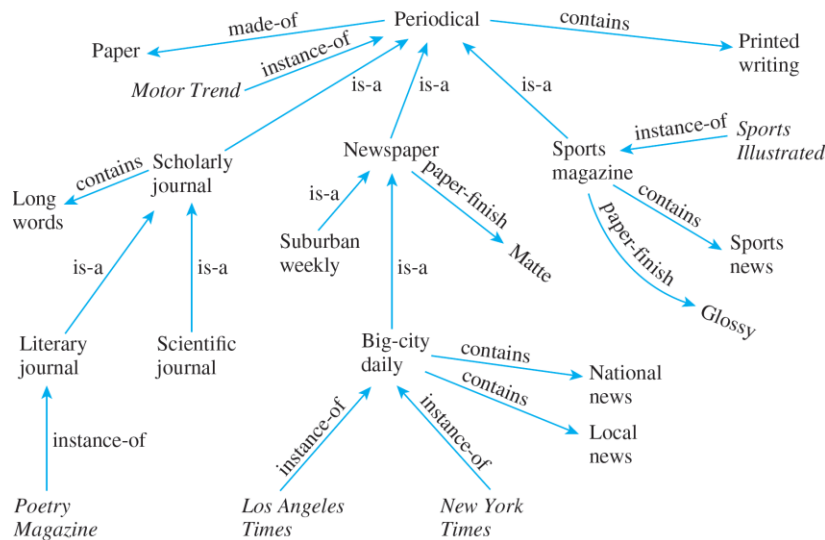


Figure 1.4.2

Question: According to this knowledge base, what paper finish does the *New York Times* use?

Review Topics

- Show logical equivalence by truth table
- Show logical equivalence by existing laws
- Argument with rules of inference
- Solve puzzles by rules of inference

Example 2.1.3 – *And, Or, and Inequalities*

Suppose x is a particular real number. Let p , q , and r symbolize “ $0 < x$,” “ $x < 3$,” and “ $x = 3$,” respectively. Write the following inequalities symbolically:

- a. $x \leq 3$
- b. $0 < x < 3$
- c. $0 < x \leq 3$

Example 2.1.3 – *Solution*

- a. $q \vee r$
- b. $p \wedge q$
- c. $p \wedge (q \vee r)$

Logical Equivalence

Testing Whether Two Statement Forms P and Q Are Logically Equivalent

1. Construct a truth table with one column for the truth values of P and another column for the truth values of Q .
2. Check each combination of truth values of the statement variables to see whether the truth value of P is the same as the truth value of Q .
 - a. If in each row the truth value of P is the same as the truth value of Q , then P and Q are logically equivalent.
 - b. If in some row P has a different truth value from Q , then P and Q are not logically equivalent.

Distributive Laws

Distributive laws:

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

Use truth table to show distributive laws:

Absorption Laws

Absorption laws:

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$

Use truth table to show absorption laws:

Example of Conditional Statement

Definition

If p and q are statement variables, the **conditional** of q by p is “If p then q ” or “ p implies q ” and is denoted $p \rightarrow q$. It is false when p is true and q is false; otherwise it is true. We call p the **hypothesis** (or **antecedent**) of the conditional and q the **conclusion** (or **consequent**).

- Construct a truth table for the statement form $p \vee \sim q \rightarrow \sim p$.

Example 2.2.3 – Division into Cases

Use truth tables to show the logical equivalence of the statement forms $p \vee q \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$

Example 2.2.3 – *Solution*

p	q	r	$p \vee q$	$p \rightarrow r$	$q \rightarrow r$	$p \vee q \rightarrow r$	$(p \rightarrow r) \wedge (q \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	T	T	T	T	T
T	F	F	T	F	T	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	F
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T


 $p \vee q \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$
always have the same truth values,
so they are logically equivalent

Only If and the Biconditional

Definition

Given statement variables p and q , the **biconditional of p and q** is “ p if, and only if, q ” and is denoted $p \leftrightarrow q$. It is true if both p and q have the same truth values and is false if p and q have opposite truth values. The words *if and only if* are sometimes abbreviated **iff**.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Truth Table for $p \leftrightarrow q$

Logical Equivalence with Biconditionals

Use truth tables to show the logical equivalence of the statement forms

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\sim p \wedge \sim q)$$

Logical Equivalence with Biconditionals

Use existing logical equivalences to show the logical equivalence of the statement forms:

$$1. (p \rightarrow q) \wedge (p \rightarrow \sim q) \wedge r \equiv r \wedge \sim p$$

$$2. (p \wedge q) \rightarrow ((q \rightarrow r) \leftrightarrow \sim p) \equiv \sim p \vee \sim q \vee \sim r$$

Valid and Invalid Arguments (论证)

Definition

An **argument** is a sequence of statements, and an **argument form** is a sequence of statement forms. All statements in an argument and all statement forms in an argument form, except for the final one, are called **premises** (or **assumptions** or **hypotheses**). The final statement or statement form is called the **conclusion**. The symbol $\therefore$, which is read “therefore,” is normally placed just before the conclusion.

To say that an *argument form* is **valid** means that no matter what particular statements are substituted for the statement variables in its premises, if the resulting premises are all true, then the conclusion is also true. To say that an *argument* is **valid** means that its form is valid.

Example of argument form:

$$\begin{array}{c} p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

Two premises:

- $p \rightarrow q$
- p

Conclusion:

- q

We can use a horizontal bar to separate the premises and conclusion.

The symbol $\therefore$ denotes “therefore”.

Generalization

The following argument forms are valid:

$$\text{a. } \begin{array}{l} p \\ \therefore p \vee q \end{array}$$

$$\text{b. } \begin{array}{l} q \\ \therefore p \vee q \end{array}$$

Specialization

The following argument forms are valid:

$$\text{a. } \begin{array}{l} p \wedge q \\ \therefore p \end{array}$$

$$\text{b. } \begin{array}{l} p \wedge q \\ \therefore q \end{array}$$

Elimination

The following argument forms are valid:

$$\text{a. } \begin{array}{l} p \vee q \\ \sim q \\ \therefore p \end{array}$$

$$\text{b. } \begin{array}{l} p \vee q \\ \sim p \\ \therefore q \end{array}$$

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \therefore p \rightarrow r \end{array}$$

Transitivity

The following argument form is valid:

Proof by Division into Cases

The following argument form is valid:

$$\begin{array}{l} p \vee q \\ p \rightarrow r \\ q \rightarrow r \\ \therefore r \end{array}$$

Example of Inference

Premises: $p \vee q; p \rightarrow \sim r; s \rightarrow t; \sim s \rightarrow r; \sim t.$

Conclusion: q

Solution:

[illegible]

Puzzle Problem 1 (Exercise 2.3.37)

In the back of an old cupboard you discover a note signed by a pirate famous for his bizarre sense of humor and love of logical puzzles. In the note he wrote that he had hidden treasure somewhere on the property. He listed five true statements (a–e below) and challenged the reader to use them to figure out the location of the treasure.

- a. If this house is next to a lake, then the treasure is not in the kitchen.
- b. If the tree in the front yard is an elm (榆树), then the treasure is in the kitchen.
- c. This house is next to a lake.
- d. The tree in the front yard is an elm or the treasure is buried under the flagpole.
- e. If the tree in the back yard is an oak (栎树), then the treasure is in the garage.

Where is the treasure hidden?

Solution

1. The treasure is not in the kitchen. (*by (c) and (a) and modus ponens*)
2. The tree in the front yard is not an elm. (*by (b) and (1) and modus tollens*)
3. The treasure is buried under the flagpole. (*by (d) and (2) and elimination*)

Puzzle Problem 2 (Exercise 2.3.39)

The famous detective Percule Hoirot was called in to solve a baffling murder mystery. He determined the following facts:

- a. **Lord Hazelton**, the murdered man, was killed by a blow on the head with a brass candlestick.
- b. Either **Lady Hazelton** or a maid, **Sara**, was in the **dining room** at the time of the murder.
- c. If the **cook** was in the **kitchen** at the time of the murder, then the **butler** (男管家) killed Lord Hazelton with a fatal dose of strychnine (a very poisonous drug).
- d. If **Lady Hazelton** was in the **dining room** at the time of the murder, then the **chauffeur** (司机) killed Lord Hazelton.
- e. If the **cook** was not in the **kitchen** at the time of the murder, then **Sara** was not in the **dining room** when the murder was committed.
- f. If **Sara** was in the **dining room** at the time the murder was committed, then the **wine steward** killed Lord Hazelton.

Is it possible for the detective to deduce the identity of the murderer from these facts? If so, who did murder Lord Hazelton? (Assume there was **only one cause of death**.)

Solution

Statements:

P: Lord Hazelton was killed by a blow on the head with a brass candlestick.

Q: Lady Hazelton was in the dining room at the time of the murder.

R: Sara, the maid, was in the dining room at the time of the murder.

S: The cook was in the kitchen at the time of the murder.

T: The butler killed Lord Hazelton with a fatal dose of strychnine.

U: The chauffeur killed Lord Hazelton.

V: The wine steward killed Lord Hazelton.

Statement Forms:

(a) P

(b) $Q \vee R$

(c) $S \rightarrow T$

(d) $Q \rightarrow U$

(e) $\sim S \rightarrow \sim R$

(f) $R \rightarrow V$

(g) Assume there was only one cause of death: $\sim (P \wedge T)$

Equivalences:

(h) $\sim (P \wedge T) \equiv \sim P \vee \sim T$ by De Morgan's Laws

(i) $\sim (\sim P) \equiv P$ by negation

Arguments:

1. $\sim P \vee \sim T$ by (h)
 $\sim (\sim P)$ by (i)
 $\therefore \sim T$ by elimination
2. $S \rightarrow T$ by (c)
 $\sim T$ by the conclusion of (1)
 $\therefore \sim S$ by modus tollens
3. $\sim S \rightarrow \sim R$ by (e)
 $\sim S$ by the conclusion of (2)
 $\therefore \sim R$ by modus ponens
4. $Q \vee R$ by (b)
 $\sim R$ by the conclusion of (3)
 $\therefore Q$ by elimination
5. $Q \rightarrow U$ by (d)
 Q by the conclusion of (4)
 $\therefore U$ by modus ponens

U: The chauffeur killed Lord Hazelton.